🇨🇷 Costa Rica — Substation ESG Report v4.2

Substation-level Environmental, Social, and Governance assessment aligned to CSRD/ESRS, EU Taxonomy, TCFD, SFDR, and UN PRI frameworks, across the 169-substation Costa Rica fleet. Updated monthly via the SSI Index v4.2 scoring engine (101 variables · 34 sources · 11 modifiers · Re composite bounded [0.920, 1.787]).

SUBSTATION

Subestación La Caja A

w557557072 · San José, san-jose

SSI SCORE (R_{MEDIAN})

0.473

● Medium Risk 12.4th percentile

VOLTAGE CLASS

138 kV

High-voltage transmission

CONFIDENCE INTERVAL

0.414 – 0.531

P5–P95 · CI width 0.117 · medium confidence

MARKOV ETTC

35 yr

Expected time to criticality

DER PENETRATION

0.02

T1 transition stress score

Component Fingerprint + Re composite v4.2

Six-component weighted decomposition of the SSI composite score (C / V / I / E / S / T) — the substation's structural risk profile — PLUS the v4.2 **Re composite**, a bounded resilience aggregate of the new modifier family (R6c flood + R6d wildfire + R6e winter + R8 adapt + R9 compound + R10 just). Layout: *left* — the 6 components rendered as a weighted contribution bar · *right* — the same 6 components rendered as horizontal bars side-by-side with Re_norm as the comparable 7th bar (terracotta) on the [0, 1] scale · *below* — the Re composite tier card (numeric value + bounded trajectory bar + classification + formula explainer). Re composite re-appears as the 7th axis of the ESG Radar in Section A below.

WEIGHTED CONTRIBUTION TO R_{BASE}

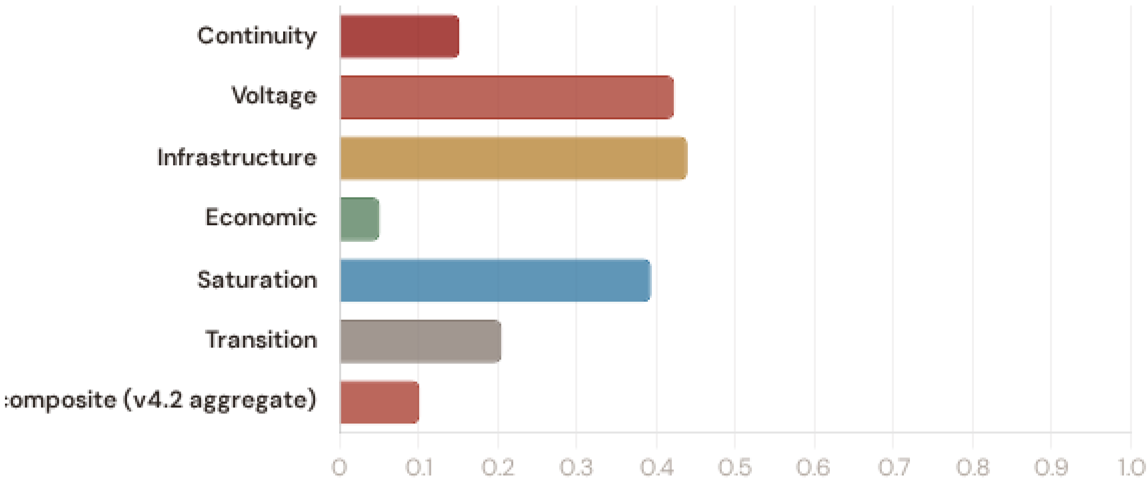
V4.2 RE COMPOSITE (AGGREGATE, NOT IN R_{BASE})

low exposure

Continuity **0.152** ×0.3 · Voltage **0.423** ×0.1 · Infrastructure **0.439** ×0.25 · Economic **0.051** ×0.1

Saturation **0.394** ×0.2 · Transition **0.205** ×0.05 · **Re composite** v4.2 aggregate **0.102** agg

R_{base} **total** (weighted sum) **0.292** Σ = 1.00



0.102

RE_NORM

LOW EXPOSURE

LOW (0.00) · MODERATE (0.25) · ELEVATED (0.50) · (1.00)

Re_raw 1.1540 (bounds [0.920, 1.787]; normalised to [0, 1])

FORMULA

```
Re_raw = (R6d × R6e × R8 × R9 ×
R10) + (R6c − 1.00)
```

Bounded resilience aggregate of the v4.2 family. Re < 1.0 when R8 adaptive capacity dominates other near-neutral factors.

Re composite tier classification: **low** (Re_norm < 0.25) · **moderate** (0.25–0.50) · **elevated** (≥ 0.50). Bounds [0.920, 1.787] map to Re_norm [0, 1] via clip((Re_raw - 0.920) / (1.787 - 0.920), 0, 1).

Markov Degradation Profile

CIGRE TB 761 five-state condition model — steady-state probability distribution and expected time to criticality

STEADY-STATE DISTRIBUTION

GOOD	AGED	DEGRADED	CRITICAL
0%	0%	0%	0%
$\pi1$ = 0.000	$\pi2$ = 0.000	$\pi3$ = 0.000	$\pi4$ = 0.000

Markov 5-state model calibrated from CIGRE Technical Brochure 761 (2019). States represent asset condition from new/good through critical. Steady-state probabilities indicate the long-run proportion of time the asset spends in each condition state given current operating environment.

RISK SCORE

28.200

Composite Markov risk

ETTC

35 yr

Expected time to critical

P(CRITICAL) 20YR

16.9%

Probability of reaching critical

CORROSION

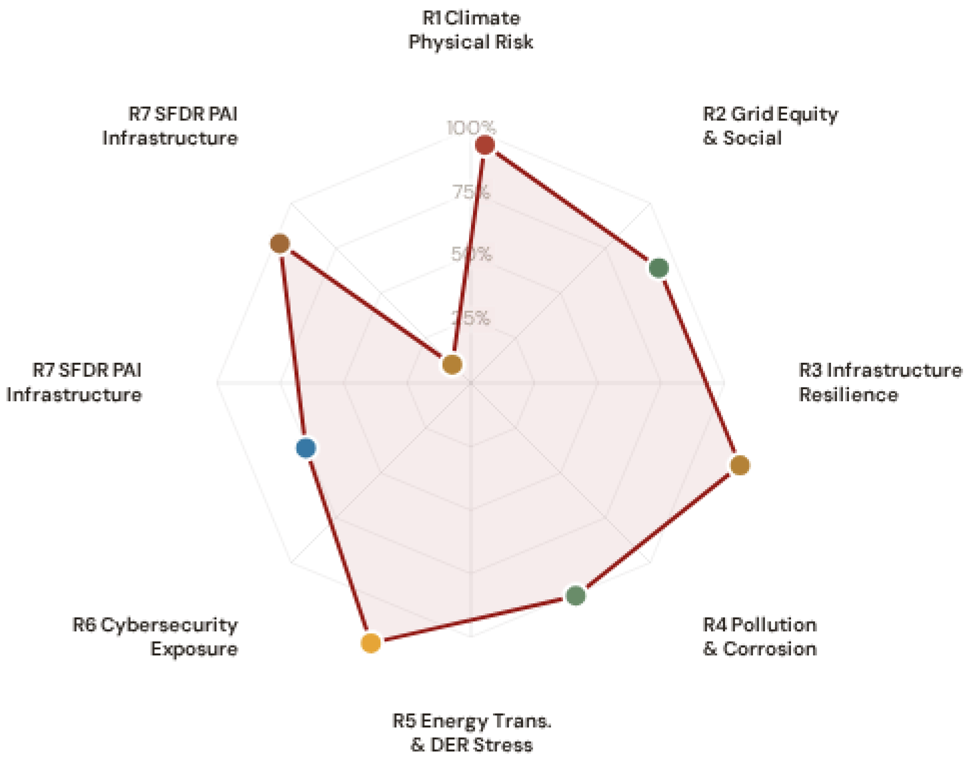
C3

ISO 9223 classification

ESG Radar — Seven-Report Overview v4.2

A

Composite readiness across 7 ESG dimensions per FC v3 §14 (R1 Climate Physical · R2 Grid Equity & Social · R3 Infrastructure Resilience [Re composite ESG home] · R4 Pollution & Corrosion · R5 Energy Transition & DER Stress · R6 Cybersecurity · R7 SFDR PAI Infrastructure, FC v3 §14 subsection 13.7), each linked to UN Sustainable Development Goals. *Note: R7 here is the SFDR ESG-disclosure axis (Re_norm proxy under Convention #7) — distinct from the v4.0.2 formula-chain R7 cyber/digital modifier documented in methodology.html. The two R7s share a number by design (V4_2_IMPLEMENTATION_ARCHITECTURE.md §4.5).*



Re composite v4.2 7th axis · **Re_raw** 1.1540 · **Re_norm** 0.102 · LOW EXPOSURE

- **R1 · Climate Physical Risk Assessment** READY
  SDG 13 Climate Action · SDG 9 · SDG 11
  SDG 13 SDG 9 SDG 11

- **R2 · Grid Equity & Social Vulnerability** READY
  SDG 7 Affordable and Clean Energy · SDG 10 · SDG 1
  SDG 7 SDG 10 SDG 1

- **R3 · Infrastructure Resilience [Re composite home]** READY
  SDG 9 Industry, Innovation, Infrastructure · SDG 13 · SDG 12
  SDG 9 SDG 13 SDG 12

- **R4 · Pollution & Corrosion** READY
  SDG 11 Sustainable Cities and Communities · SDG 3 · SDG 15
  SDG 11 SDG 3 SDG 15

- **R5 · Energy Transition & DER Stress** READY
  SDG 7 Affordable and Clean Energy · SDG 13 · SDG 9

SDG 7 SDG 13 SDG 9

- R6 · Cybersecurity Exposure PARTIAL

  SDG 9 Industry, Innovation, Infrastructure · SDG 16

  SDG 9 SDG 16

- R7 · SFDR PAI Infrastructure Disclosure READY

  SDG 12 Responsible Consumption and Production · SDG 9 · SDG 13

  SDG 12 SDG 9 SDG 13

- R7 · SFDR PAI Infrastructure READY v4.2

  SDG 9 Industry, Innovation, Infrastructure · SDG 13 · SDG 11 · Re_norm = 0.102 (low exposure)

  SDG 9 SDG 13 SDG 11

C Audit Trail & Methodology Note

Traceability for CSRD Article 29a limited assurance

ITEM	VALUE
Scoring Engine	SSI Index **v4.2** — Gaussian copula Monte Carlo (10,000 iterations) · 11 modifiers (5 baseline + 6 v4.2 resilience family) · Re composite bounded [0.920, 1.787]
Weight Architecture	C=0.30, V=0.10, I=0.25, E=0.10, S=0.20, T=0.05
Modifiers Applied	R3 C_mult: 1.029, R4 F_topo: 1.000, R6 restoration: 0.900, R6 seismic: 1.250, R6 volcanic: 1.000, R6 hurricane: 0.020, R6 hydro_deficit: 0.080, R7 cyber: 1.025, R10 just: 1.021, R6d wildfire: 1.052, R6c flood: 1.055, R8 adapt: 0.980, R9 compound: 1.038, R6e winter: 1.006, R7 cyber_v2: 1.014
Degradation Model	Markov 5-state, CIGRE TB 761 calibration
Thermal Model	IEEE C57.91 transformer loading curves
Report Date	17 March 2026
Reporting Period	Calendar year 2025
Refresh Cadence	Annual (data ingestion → scoring engine → display)
Substation	Subestación La Caja A (w557557072)
Data Assurance Level	All inputs from institutional open-source with citable vintage
Re composite (v4.2)	Re_raw = 1.1540 · Re_norm = **0.102** · **LOW** exposure · bounds [0.920, 1.787]

D

Re composite — bounded resilience aggregate (v4.2)

Resilience composite anchoring CSRD/ESRS climate-resilience disclosure

The Re composite (v4.2) is a bounded resilience aggregate capturing the new R6c flood / R6d wildfire / R6e winter / R8 adapt / R9 compound / R10 just modifier family in a single scalar suitable for CSRD/ESRS climate-resilience disclosure:

```
Re_raw = (R6d × R6e × R8 × R9 × R10) + (R6c − 1.00)
```

Bounds: **[0.920, 1.787]**. Re_raw < 1.000 when R8_adapt < 1.00 dominates other near-neutral factors. Costa Rican fleet observed range across 169 substations: the methodology-bounded extremes [0.920, 1.787] — low-end values arise at interior low-exposure substations with R8_adapt < 1.00 dominating near-neutral other modifiers, high-end values at coastal compound-exposure substations where R6c flood + R9 compound stack multiplicatively. v4.2 explicitly retires the pre-Path-B "winter exposure is a real climate burden" reading per Convention #6 §2.3.1 (Bourgouin freezing-rain p99 tail-event amplifier per §6 design — winter exposure signal correctly lives in v4.0.2 I1 Snow/Ice metric, where it methodologically belongs). See W1-W10 FC v2 for per-substation worked examples + W4 Climate resilience baselines.

Foundation contact

The SSI Index is a non-profit foundation project, please mail to ssi_index@ikenga.eu for more details.